

EBOOK

When Every Second Counts:

A guide to ransomware remediation
for security professionals

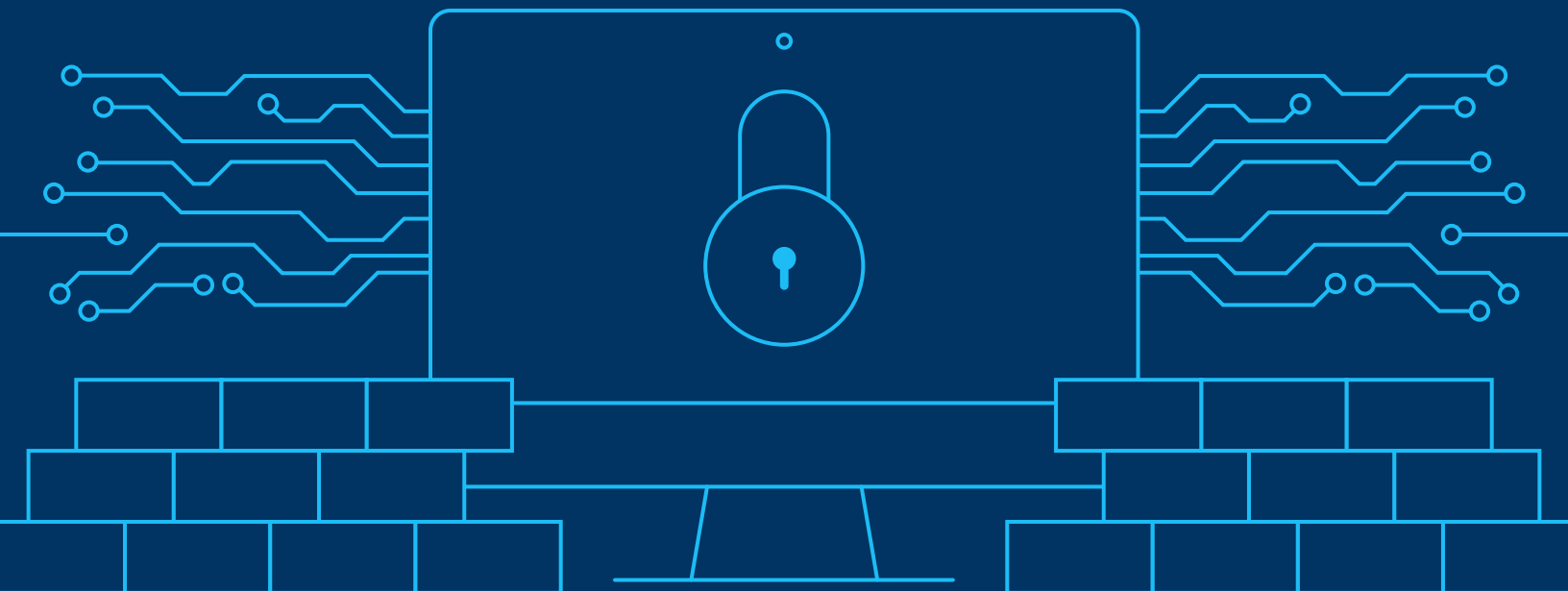
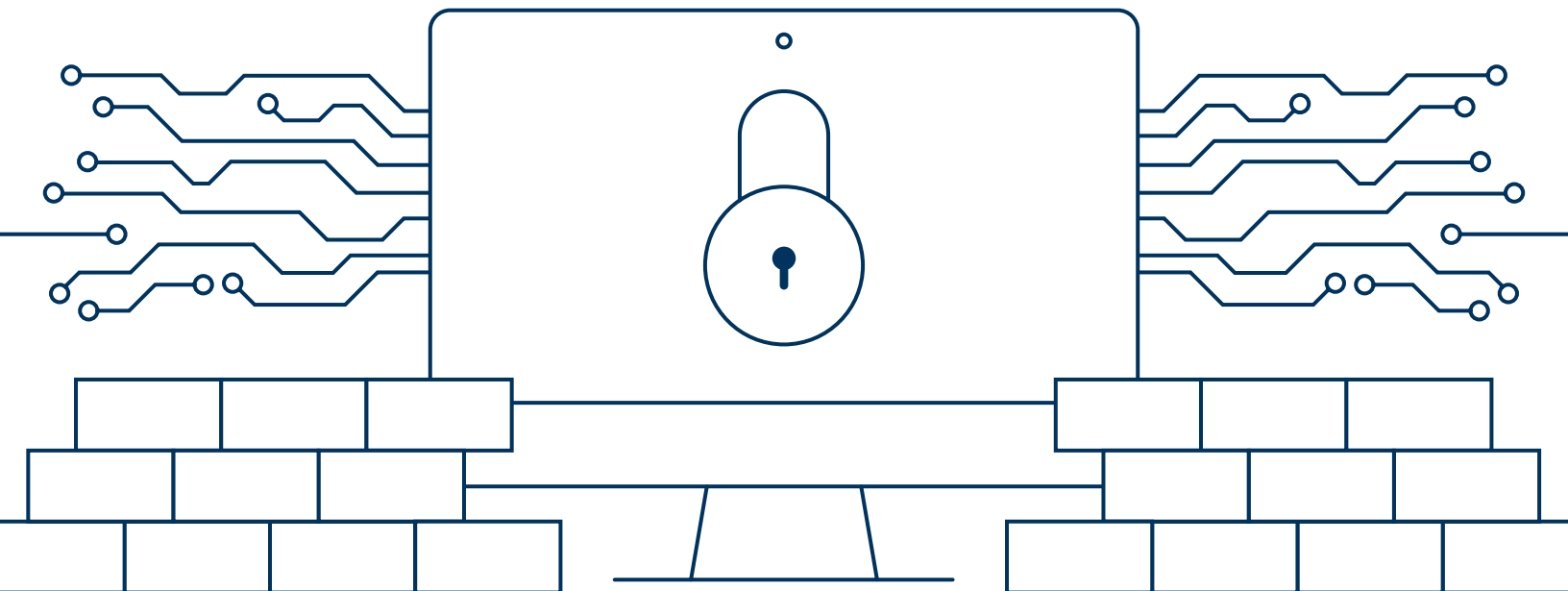


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Preparing for ransomware attacks

As more organizations adopt digital business models, data has become a lucrative target for cyberattacks. Even with defense mechanisms, ransomware attacks continue to capture organizations' data successfully.

According to a survey by Sophos, the average ransom payment has been steadily increasing over the years, reaching around **\$1.5 million** in 2023.¹ However, the ransom payment is only a part of the total cost of a successful attack—the same survey found that when you add downtime, labor, device and network costs, lost opportunities, etc., the estimated mean recovery cost was as much as **\$1.82 million**, excluding any ransom paid.¹

An enterprise data backup and recovery strategy could be an essential foil against data loss. Today, however, backup data is increasingly a target for ransomware, meaning that your last line of defense could become compromised. Many legacy backup systems have an outdated architecture that can make software, servers, and repositories more exploitable to attacks. Furthermore, legacy backup systems typically can't determine if data has been encrypted, so they may risk backing up the compromised data and unintentionally prolonging the attack.

Despite your best efforts, your organization could still be hit with a ransomware attack. If so, it'll face an uncomfortable decision: Do you pay the ransom and hope you can get your data back, or do you use your recovery plan to save your data? In these moments, speed is critical—you don't want to waste precious time weighing the pros and cons of giving in to a cybercriminal's demands.



¹The State of Ransomware 2023 | Sophos

BARRIERS TO RECOVERY

Why doesn't paying ransom guarantee data recovery? When you pay the ransom, the attackers will (allegedly) give you a decryptor to recover the data they encrypted. But there are a few problems:



They don't always provide the decryptor.



The decryptor can be slow to recover data.



The decryptor doesn't always work.



They may have used "wiperware," rendering decryptors useless.



Is paying the ransom worth it?

- Only **35%** of those affected by ransomware recovered all of their data after an incident²
- **80%** of those who paid the ransom got hit with a second attack³
- **68%** of organizations who paid the ransom reported they were hit again within a month—for a higher ransom³

Studies show that paying a ransom does not guarantee the safe return of your data. Indeed, many companies that pay a ransom get hit with a second attack—sometimes almost immediately.³ Recovering after a ransomware attack is painful and time-consuming. After recovering from a ransomware attack, the last thing you want to do is go through it all again. *How do you shorten the recovery time and inoculate yourself against a future attack?*

REFUSE AND REMEDIATE

The best course of action is to refuse to pay. However, you should have a strong remediation plan in place that lets you recover quickly and eliminate the vulnerabilities that made the attack possible in the first place. This guide will help you build a ransomware remediation plan that optimizes those precious minutes and hours when an attack occurs. It includes:



Six preparatory steps you can take today



Tips for selecting a robust data security solution



A three-phase approach to dealing with an attack when it happens



Top questions for vetting data security vendors

²The 2023 Global Ransomware Report | Fortinet

³Ransomware The True Cost to Business 2022 | Cybereason

What you can do now

Six steps to prepare for an attack

No matter how many defenses you put up, the fact is that most organizations will suffer from a ransomware attack. A recent survey revealed that of the 78% of organizations that said they were “very” or “extremely” prepared to mitigate an attack, 50% still fell victim to ransomware.²

Rather than hoping your organization will never be targeted, it’s best to behave as if an attack is imminent and prepare accordingly. Here are some basic steps you can take today to make recovery as fast and painless as possible.



STORE DATA IN AN IMMUTABLE FORMAT: Storing data in an immutable format means it cannot be altered, modified, or deleted once it is written or recorded. The data becomes read-only and remains in its original state indefinitely.



REQUIRE CONTINUOUS HUMAN SECURITY TRAINING: Today, **69%** of ransomware attacks come from email phishing—still the most common point of access for ransomware attacks.⁴ With human error posing such a vulnerability, you must schedule and perform quarterly training to help staff spot attempted attacks. Additionally, institute a patch management program to fix known vulnerabilities in outdated software.



INCREASE THE FREQUENCY OF SENSITIVE DATA ACCOUNTING ACROSS DATACENTERS: Before data theft attacks, organizations typically account for their sensitive data every quarter. Since data attacks are growing, organizations should account for all data types across all platforms daily.



ESTABLISH A RECOVERY TEAM: Delegate different recovery tasks to specific team members and ensure they have the skills and tools to handle those tasks. Create a cross-functional operation between the recovery and data investigation (security) teams to ensure smooth communications during an attack.



CATEGORIZE AND PRIORITIZE APPLICATIONS BASED ON BUSINESS IMPACT: Assess the potential business impact of an attack on different applications, so you know which ones to prioritize in the event of an attack. Additionally, identify the application owners who can quickly validate functionality and data integrity.



ESTABLISH RESILIENT AUTHENTICATION AND AUTHORIZATION CONTROLS: Implement layers of security, such as strong passwords (regularly updated), multi-factor authentication (MFA), least privileged access, and roles-based access controls.

⁴ 2023 Ransomware Insights | Barracuda Networks

Checklist: What to do when ransomware attacks

Taking these preparatory steps will help your organization recover quickly in the event of a ransomware attack. The key to minimizing the damage is keeping your team calm, connected, and focused on their tasks as you progress through your remediation plan.

Picture this: Even after your diligent preparation, an attack still happens. Someone in your organization clicks on a convincing (but bogus) email offer, unleashing a ransomware attack. The attackers send a ransom note informing you that your data has been encrypted and demanding payment for the decryptor.

This three-phase plan will help your organization recover faster and strengthen your data security posture.

PHASE 1: ACTIVATE YOUR RESPONSE PLAN

Most ransomware attacks are detected only in their final stage—after attackers have encrypted data and interfered with business processing, resulting in a flood of calls to the help desk. In these cases, organizations need to act immediately to prevent further damage.

- ✓ **CHECK RETENTION TIME AND PAUSE EXPIRATION:** Short retention times increase the risk of backups expiring before the completion of remediation efforts. Extending retention times will help ensure recovery to the most recent clean state prior to the attack. Additionally, snapshot expiration should be carefully reviewed to ensure no valid snapshots expire, which would affect data recovery.
- ✓ **ALERT THE AUTHORITIES:** Certain organizations may be legally required to notify authorities and users. This is highly dependent on your business, industry, and location. Inform law enforcement, customers, and other authorities as necessary following an attack.
- ✓ **ISOLATE THE COMPROMISED DEVICE AT THE NETWORK LEVEL:** Your first impulse might be to shut down the system. However, isolating is a better option by disconnecting all compromised devices from the network and any shared storage. Additionally, any volatile system data held in temporary memory should be preserved for evidence collection and indicators of compromise analysis.
- ✓ **ENSURE BACKUPS HAVEN'T BEEN COMPROMISED:** This will help you identify a safe point in time for restoration. Not selecting a safe recovery point could reintroduce the ransomware if it's already in your backups.
- ✓ **CREATE A COMMUNICATION BRIDGE WITH THE AFFECTED PARTIES:** Key members of the organization must be able to share information during an attack to mitigate damage and coordinate the response. However, this process is often ad hoc in many organizations. It is essential to understand who is responsible for what system and how they plan to share information in a crisis.

PHASE 2: IDENTIFY THE SCOPE OF THE ATTACK

Knowing exactly what data was compromised will help the response team focus resources on the point of attack. But you must discover where the malware is, what data has been encrypted, and whether sensitive data has been affected.

- ✓ **DIAGNOSE MALWARE, ENCRYPTION, AND SENSITIVE DATA:** You want to make the most of your limited investigation time and resources by focusing only on impacted files. Determine if the attack includes encryption and/or data exfiltration. Then, determine the blast radius of encryption and generate a sensitive data report (to be delivered to Legal). Lastly, investigate compromised systems for malware indicators.
- ✓ **PERFORM THREAT HUNTING ON BACKUPS:** You must have a process for validating your restoration points to ensure they are free from malware. Scan your backup snapshots for indicators of compromise—such as patterns, hashes, or rules that can help identify malicious files or activities. This will help avoid reinfection and ensure your data's integrity and availability.
- ✓ **PERFORM RECOVERIES:** Pinpoint clean points in time that don't contain malware so you can restore backups without reintroducing the threat.

PHASE 3: DOCUMENT LESSONS LEARNED

Remember: Many businesses that are hit once will become repeat victims. Recovering from an attack should include fortifying your security posture to avoid going through the same situation a second time.

- ✓ **CONTINUOUSLY MONITOR FOR TRACES OR INDICATORS OF COMPROMISED DATA:** Keep your threat intelligence sharp by constantly scanning data for anomalies or indicators of malicious activity.
- ✓ **DOCUMENT THE SECURITY GAPS THAT WERE EXPLOITED:** Attackers will try to make money by selling the credentials that got them into your system in the first place. Discover exactly how and where the malware was initially introduced so you can block off that avenue for good.
- ✓ **STRENGTHEN SECURITY TO PREVENT A RECURRENCE OF THE ATTACK:** Whether you need to increase employee security training, boost visibility into network traffic, or strengthen access controls, take the necessary steps to enhance your security posture so the same attack does not re-victimize you.



Tips for selecting the best data security solution

Continuous monitoring of critical IT systems is a resource-intensive process, and few organizations invest in the data auditing capacity required to alert them to all vulnerabilities in real time. Instead, a robust data security solution will let you constantly keep tabs on sensitive data so you can respond quickly if an attack happens.

What are the hallmarks of a robust data security solution?

STREAMLINED RECOVERY AT SCALE

To prevent business disruption, it's essential to swiftly identify and restore your data to its latest clean version—whether for a full or partial system recovery. Quickly accessible backup data supports quick recovery without rehydration (the process of decompressing and importing backup data), while API-driven automation enhances flexibility and accelerates large-scale search and recovery efforts.

DATA STORED IN A NATIVE IMMUTABLE FILESYSTEM

Select a data security vendor that stores all data in an immutable format. Immutability means that once data has been ingested, modification of the data is not possible. Deletion is only supported for data marked as expired (i.e., garbage collection). Backup data should never be available for an external client to modify at any time, as this easily opens up that data to being corrupted or deleted by an attacker.

GRANULAR IMPACT DIAGNOSIS

During an attack, the manual process of assessing the affected surface area typically involves sifting through millions of files to pinpoint the breadth of the attack. This can take weeks—and most businesses resort to mass restores of the entire environment, including uncompromised data, to avoid further delays. Technologies that automate the assessment of an attack's impact and provide a clear view into what applications and files were encrypted—and where those reside—help IT teams quickly restore at a more granular level. This minimizes the risk of data loss associated with mass restores.



MULTI-LAYERED DEFENSE WITH BEHAVIORAL-BASED DETECTION

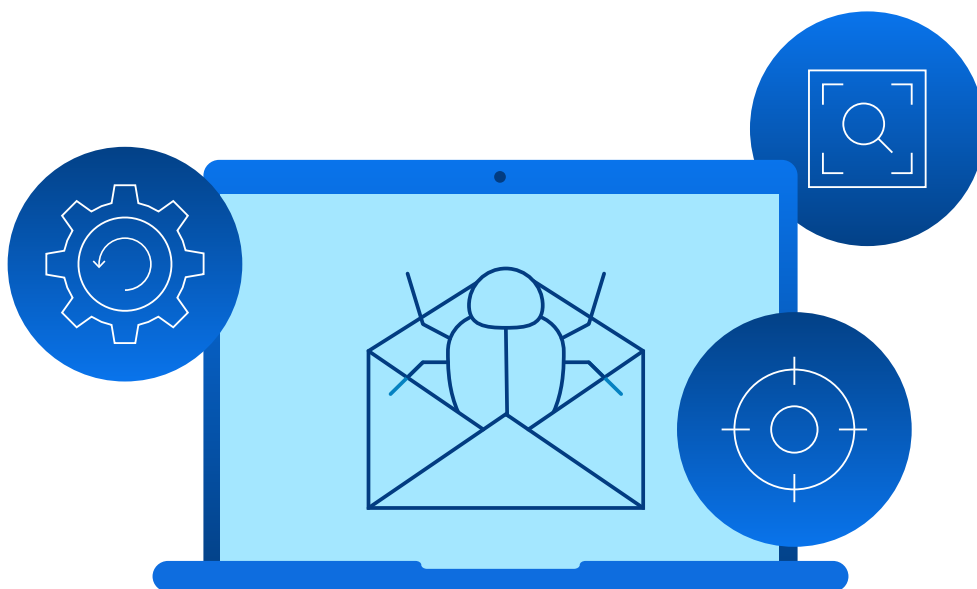
It only takes one employee to click on a phishing email or one missed patch to cause a breach. It's critical to know about a breach as soon as possible when it occurs. This means that your data needs as many eyes on it as possible. Data protection solutions—such as anomaly detection and data analysis—must be able to monitor for malware and give organizations a leg up on ensuring every block and byte are accounted for. When unusual behavior, such as ransomware, is detected, IT teams should get an immediate alert to investigate and accelerate recovery if needed.

When your system is compromised, you must swiftly understand the scope of damage. Ransomware attackers specifically cause random damage to make recovery harder. Some data security vendors use signature-based detection, which compares the characteristics of known malware variants to files on your network. However, most ransomware attacks use morphing and code obfuscation with a zero-day signature, which means that signature-based detection will only be valid after the first attack. Smart organizations can use machine learning tools to map an attack's encryption behavior, fast-tracking the response's analysis phase and allowing customers to immediately focus on threat hunting and recovery.



Trusted data security for the modern threat landscape

Cyberthreats are becoming more sophisticated—and more frequent. A robust data security solution can expedite many of the actions in this checklist, helping you get your business back on track as quickly as possible. [Explore Rubrik data security solutions](#)



Key questions for security vendors

Remember that time is your most valuable commodity when it comes to surviving a ransomware attack.

Here are three key questions to ask your vendor to ensure you can move fast during an attack.

QUESTION #1: Does your solution use a native, purpose-built platform for data protection?

The answer should be...

YES. The solution is a native, purpose-built platform for data protection that doesn't rely on third-party integrations for any of its capabilities.

FOLLOW-UP QUESTIONS: How did your platform get to be purpose-built? Do we have to set it up ourselves, or will you help us do that?

QUESTION #2: Does the data protection solution have built-in capabilities to detect encrypted files and malware automatically?

The answer should be...

YES. The best data security solution should be able to identify when the ransomware software was introduced and which systems and data were compromised. This will help you ascertain the status and impact of the situation as quickly as possible.

FOLLOW-UP QUESTIONS: Can the solution detect ransomware-based anomalous encryption? Can the solution identify sensitive data targeted by data-theft cyber campaigns? Can it identify user access to sensitive data in near real time? Can the solution support file hash or YARA rule searching to identify malware in backups?

QUESTION 3: Does the solution have the automation and the performance to handle recoveries at scale?

The answer should be...

YES. Anything you can do to speed up recovery will help minimize the damage. The best data security solution will use proven and tested automation to recover NAS systems, virtual environments, database servers, and file sets across multiple servers.

FOLLOW-UP QUESTIONS: What ML-based technologies does your solution include? Can it deliver near-zero recovery time objectives (RTOs) for VMs, file shares, and databases?

Choose complete cyber resilience

Rubrik is a cybersecurity company with a mission to secure the world's data. We pioneered Zero Trust Data Security™ to help organizations achieve business resilience against cyberattacks, malicious insiders, and operational disruptions.

Built with a zero trust design and powered by machine learning, Rubrik Security Cloud delivers complete cyber resilience in a single platform across enterprise, cloud, and SaaS. Our platform automates data policy management and enforcement, safeguards sensitive data, delivers data threat analytics and response, and orchestrates rapid cyber and operational recovery.

Cyberattacks are inevitable. Increasing your cyber resilience with a data security solution like Rubrik Security Cloud helps reduce your chances of being victimized and minimize the monetary and time loss when an attack occurs.



Trusted cyber resilience

Rubrik is trusted by over 5,500 leading organizations around the world. [Read customer stories](#)

- **St. Luke's University Health Network** deployed Rubrik to protect millions of patient records and will see **73%** TCO savings over three years.
- **The Home Depot** deployed Rubrik to 2,200 remote sites and saved **\$1 million** annually through consolidation.
- **Carhartt** consolidated onto Rubrik's data security solution for cloud and on-premises, resulting in over **50%** monthly cost savings.
- **Payette** used Rubrik Cloud Data Protection to help thwart an ongoing cyberattack, getting critical systems back online in **less than 24 hours** and fully recovering all data targeted in the attack.

"We want to make sure we can recover our systems and technology, but we can also make sure we can recover them reliably, securely, and completely."

—Magesh Bagavathi, SVP and Global CTO, [PepsiCo](#)

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